

Women in Movies: Representation and Success

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Website Presentation: <https://senakahya.github.io/Sena-Kahya-DSA210-Term-Project/>

1. Motivation

Movies are both cultural products and commercial products. They entertain audiences, but they also shape how different groups are represented in popular culture. The roles given to women, the visibility of female characters, and the presence of women behind the camera can influence which stories are told and whose perspectives become central. This makes gender representation in film an important topic not only socially, but also analytically.

The motivation for this project comes from the question of whether female representation is only a matter of visibility, or whether it is also connected to how movies perform. Discussions about representation often stay at a general level, but movie success itself is not simple. A movie can be highly rated without reaching a large audience, or it can be commercially successful without receiving especially high ratings. For that reason, this project separates success into three outcome groups: ratings, audience engagement, and financial performance.

The main objective of this project is to examine how female representation on screen and behind the camera is associated with these different movie outcomes. On-screen representation is measured through variables such as female cast share and female lead status. Behind-camera representation is measured through variables such as female director status, female crew ratio, and female creative roles.

The project is guided by four research questions:

- Do movies directed by women differ from movies without female directors?
- Is female cast representation associated with ratings, engagement, or financial outcomes?
- Is female crew representation associated with movie success?
- Do representation variables improve predictive models of movie success?

Overall, this report aims to understand whether female representation carries meaningful information about movie outcomes, while avoiding a single “successful or not successful” interpretation. The goal is not to make causal claims, but to provide a data-based analysis of how representation and different forms of movie success are related.

2. Data Sources and Data Preparation

The datasets used in this project were obtained from Kaggle. The main source is The Movies Dataset, which provides movie-level metadata and credits information for thousands of films. It includes variables such as movie title, release date, genre, runtime, budget, revenue, popularity, ratings, and cast and

crew records. This dataset forms the base of the analysis because it contains both movie success measures and the cast/crew information needed to construct gender representation variables.

To make the analysis more complete, two additional Kaggle datasets were added to the main movie dataset. The first is a Bechdel/IMDb dataset, which provides Bechdel ratings, Bechdel pass status, IMDb average ratings, and IMDb vote counts. These variables add both a story-based representation measure and an additional rating platform. The second additional source is an Oscar awards dataset, which provides nomination and win information. These records were matched to movies mainly by title and release year.

The full cleaning and merging process is documented in notebooks/01_data_cleaning_movie_dataset.ipynb. During data preparation, movie identifiers were standardized so that records from different sources could be matched more reliably. MovieLens ratings were aggregated at the movie level, and cast and crew records were processed into representation features. After this step, the Bechdel/IMDb and Oscar data were merged into the main movie table.

The final cleaned dataset contains 45,346 movies. One important cleaning decision concerns the financial variables. In the raw metadata, many movies have budget or revenue values recorded as zero. These values were treated as missing rather than real zeros, because they usually indicate that financial information was not available. For this reason, financial analysis is based only on movies with positive budget and positive revenue values.

Dataset component	Movies / records	Use in analysis
Final cleaned dataset	45,346 movies	Main analysis base
IMDb / Bechdel coverage	7,709 movies	IMDb ratings, votes, and Bechdel status
Complete financial subset	5,375 movies	Revenue, profit, ROI analysis
Movies with credits	40,318 movies	Cast and crew representation features

Table 1. Main data coverage after cleaning and merging.

The final dataset contains both representation variables and outcome variables. The main representation features are `female_director`, `female_ratio_top5`, `female_crew_ratio`, `female_lead`, and `bechdel_pass`. These variables capture whether women are represented in directing roles, main cast positions, broader crew composition, lead roles, and Bechdel test outcomes. The main success measures include ratings, popularity, vote counts, profit, ROI, and `log_revenue`. Log transformations were used for highly skewed measures such as revenue, popularity, and vote counts so that a small number of very large movies would not dominate the analysis.

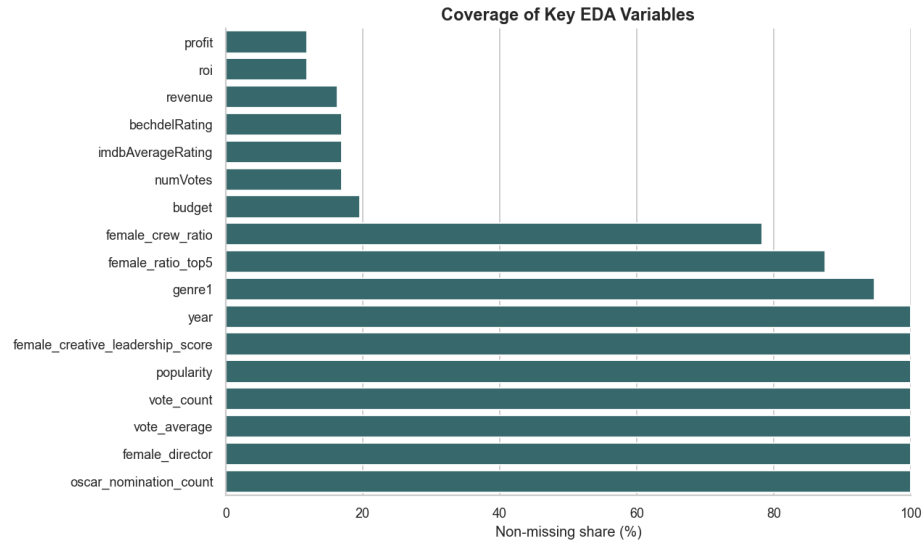


Figure 2.1. Availability of key variables used in the analysis.

3. Exploratory Data Analysis

Exploratory data analysis was used to examine how female representation appears across the cleaned movie dataset before moving into formal statistical testing. The analysis focuses on three main dimensions of representation: female directors, female presence among the top-billed cast, and female participation in behind-camera crew roles. These variables are then compared with different measures of movie success, including ratings, audience engagement, and financial outcomes.

The descriptive patterns show clear differences across time periods, genres, and production roles. Female representation is not evenly distributed across the dataset, and movies with female creative involvement often show higher female presence in other parts of production. At the same time, the relationship between representation and success is not uniform, which is why the EDA separates ratings, engagement, and financial performance instead of treating success as a single measure.

3.1. Director Representation

Female directors make up a small part of the cleaned dataset. Only about 4.1% of movies have at least one female director. The decade trend shows that this underrepresentation is not evenly distributed over time. Female director share is almost absent in the earliest decades, appears more consistently in the twentieth century, and rises more visibly around the 1990s and 2000s. Even in the most recent decades shown in the plot, female-directed movies remain a minority.

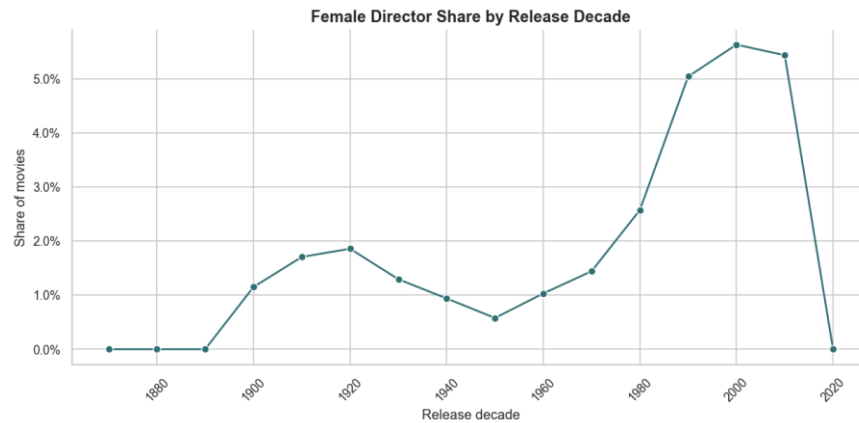


Figure 3.1. Female director share by release decade.

This time pattern is important because older and newer movies belong to different rating, audience, and financial environments. The increase in female director share is visible, but the overall level is still low. Director gender is therefore connected not only to representation, but also to historical period.

Female-directed movies also differ strongly from other movies in broader representation measures. Their average top-5 female cast share is close to half of the central cast, while movies without a female director are closer to one-third. The female lead share is also much higher, and the gap in female crew ratio is especially large.

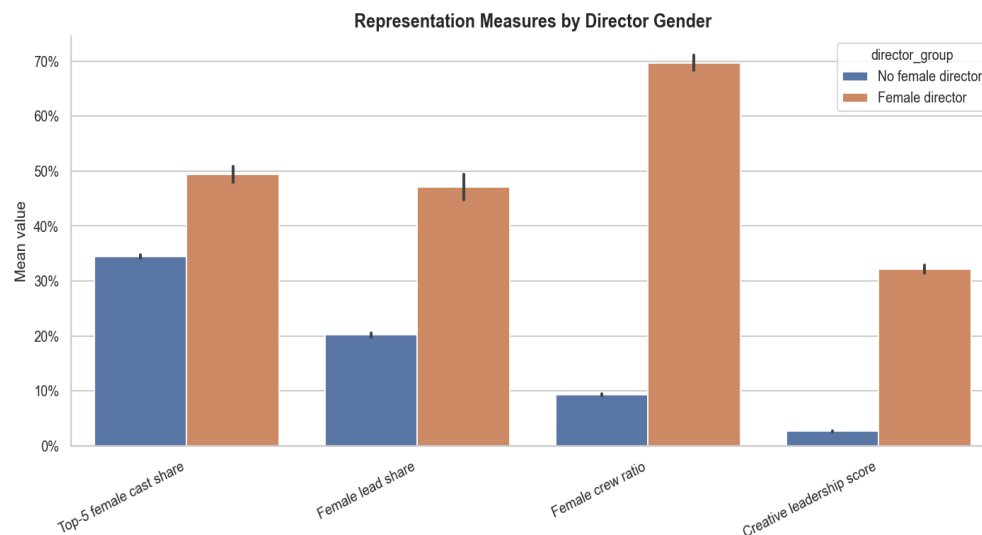


Figure 3.2. Representation measures by director gender.

This suggests that director gender is connected to the wider production structure. Movies with female directors tend to include more women in lead roles, central cast positions, crew, and creative leadership. In the descriptive data, behind-camera representation and on-screen representation move together.

Rating differences by director gender are small and mixed. Female-directed movies have a slightly higher average TMDB rating and MovieLens rating, while the IMDb rating is higher for movies without a female director. The pattern does not support a simple conclusion that one group is consistently rated higher across platforms.

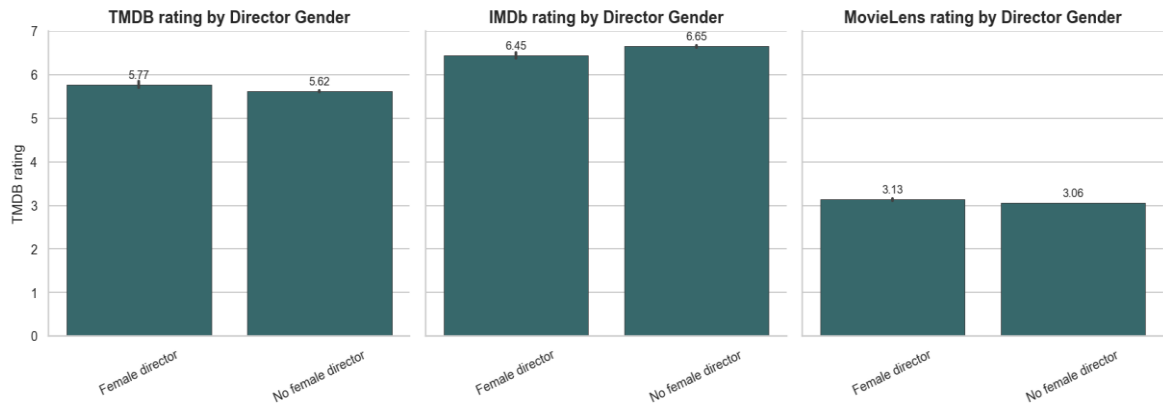


Figure 3.3. Audience ratings by director gender

Engagement metrics are also mixed, but they show a different pattern from ratings. Female-directed movies have slightly higher log popularity and log TMDB vote count, while movies without female directors have higher log IMDb vote count. Popularity and vote counts measure related but different forms of audience attention, so they do not always move in the same direction.

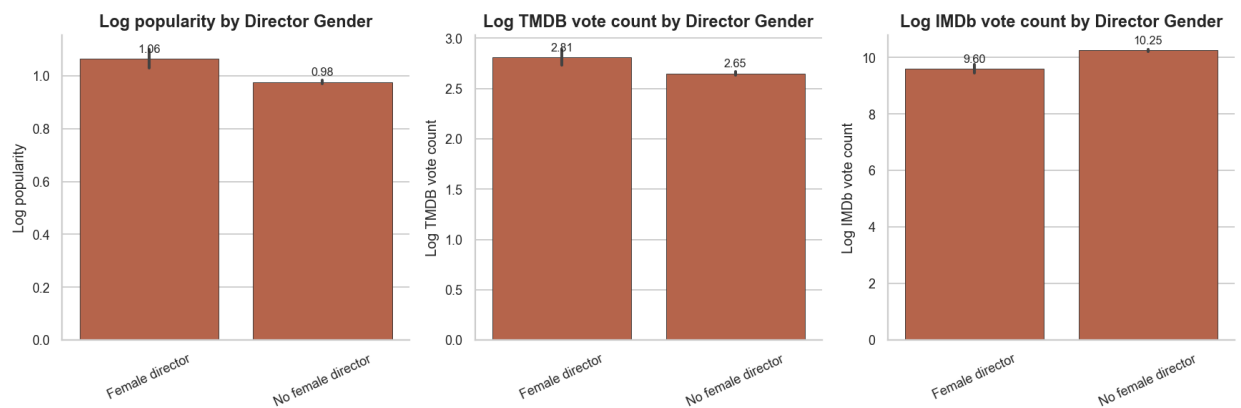


Figure 3.4. Engagement metrics by director gender

The financial comparison is based on a smaller subset: 231 female-directed movies and 5,144 movies without a female director have complete positive budget and revenue data. Within that subset, median log revenue, median profit, and median ROI are lower for female-directed movies. The difference should be read carefully because the two groups are highly imbalanced and the financial distributions are wide.

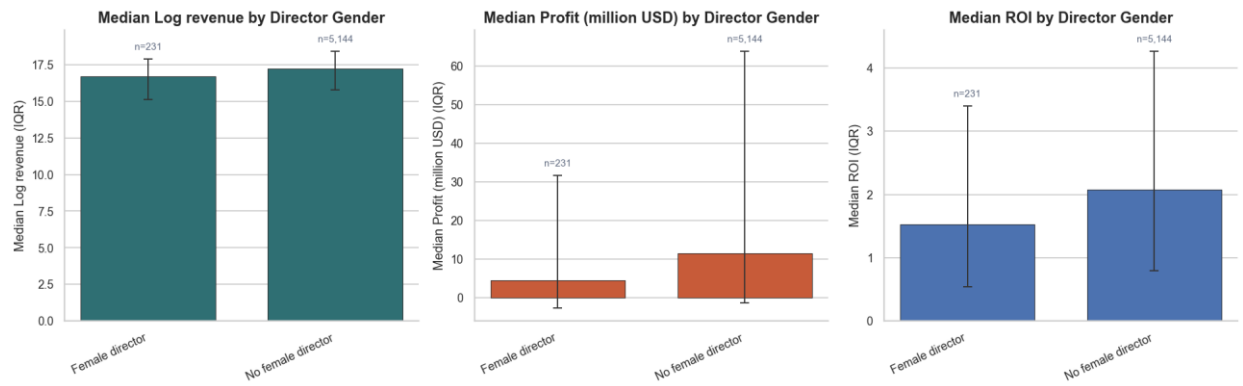


Figure 3.5. Financial outcomes by director gender.

3.2. On-Screen Representation: Top-5 Female Cast Share

Top-5 female cast share measures the proportion of women among the five most prominent cast members. Since the measure focuses on central roles, it gives a clearer view of on-screen representation than a full cast count. The distribution is uneven and clustered at discrete values because the top five positions create steps of 0%, 20%, 40%, 60%, 80%, and 100%. The median is 0.33, meaning that a typical movie has roughly one-third female representation among its most prominent cast members.

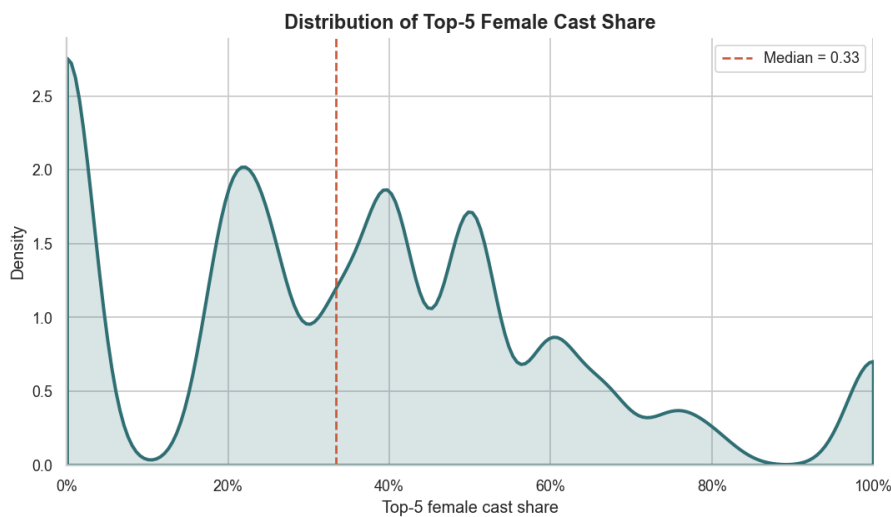


Figure 3.6. Distribution of top-5 female cast share.

Genre explains part of the variation in on-screen representation. Romance has the highest average top-5 female cast share, followed by Horror and Drama. Action and Documentary have much lower averages. This shows that female cast representation is not evenly distributed across movie types.

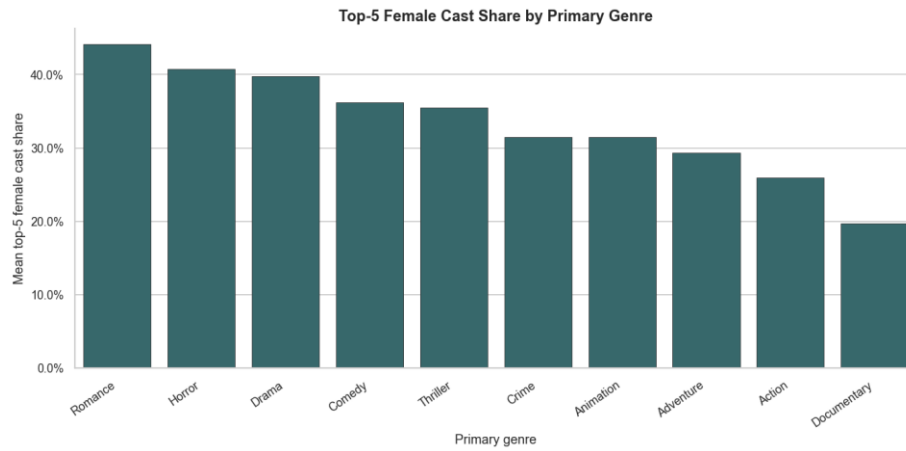


Figure 3.7. Top-5 female cast share by primary genre.

The relationship between top-5 female cast share and ratings is weak. In the notebook scatterplots, TMDb, IMDb, and MovieLens ratings all show small negative correlations with top-5 female cast share. The size of these relationships is limited, which suggests that rating outcomes are shaped by many factors beyond cast gender composition, including genre, release period, audience base, and platform-specific rating behavior.

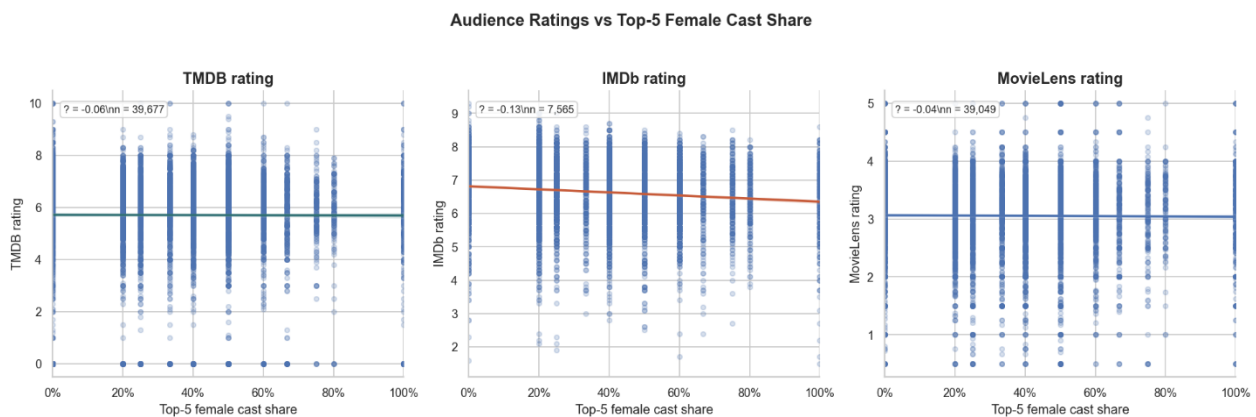


Figure 3.8. Audience ratings versus top-5 female cast share.

Engagement outcomes show a clearer descriptive shape than ratings. Log popularity is highest in the 20-40% female cast group and then decreases in the highest female cast share group. TMDb vote count follows a similar pattern, while IMDb vote count declines more steadily as the female cast share group increases. This may reflect genre and market-size differences, since movies with mostly female central casts may be less common among large franchise, action, or mass-market productions.

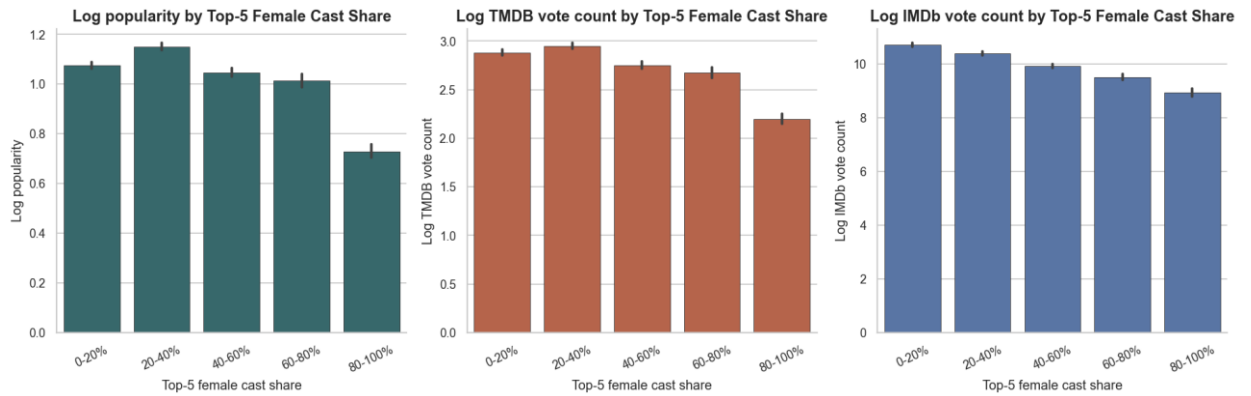


Figure 3.9. Engagement metrics by top-5 female cast share group.

The contrast between ratings and engagement is visible. A movie can receive solid ratings but attract fewer votes, while another movie can reach a large audience with average ratings. In these plots, cast representation is more visibly connected to audience reach than to rating scores, although the direction changes across engagement measures.

Financial outcomes are again limited to movies with complete positive budget and revenue. Median log revenue declines gradually as top-5 female cast share increases, and median profit is lowest in the 80-100% group. ROI is less consistent: the highest median ROI appears in the 60-80% group, while the 80-100% group still has a small sample size of only 96 movies. This means that the highest-representation category should be interpreted carefully.

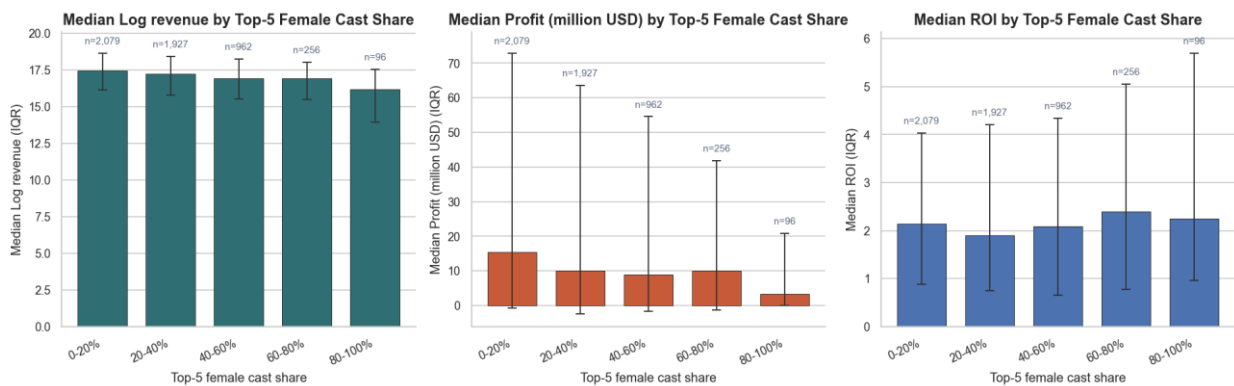


Figure 3.10. Financial outcomes by top-5 female cast share group.

3.3. Behind-Camera Representation: Crew and Creative Roles

Behind-camera representation is even more limited than on-screen representation. Across the selected roles, female participation is low: writers appear most often at 6.4%, producers at 5.7%, directors at 4.1%, editors at 4.0%, executive producers at 2.5%, and cinematographers at only 0.7%.

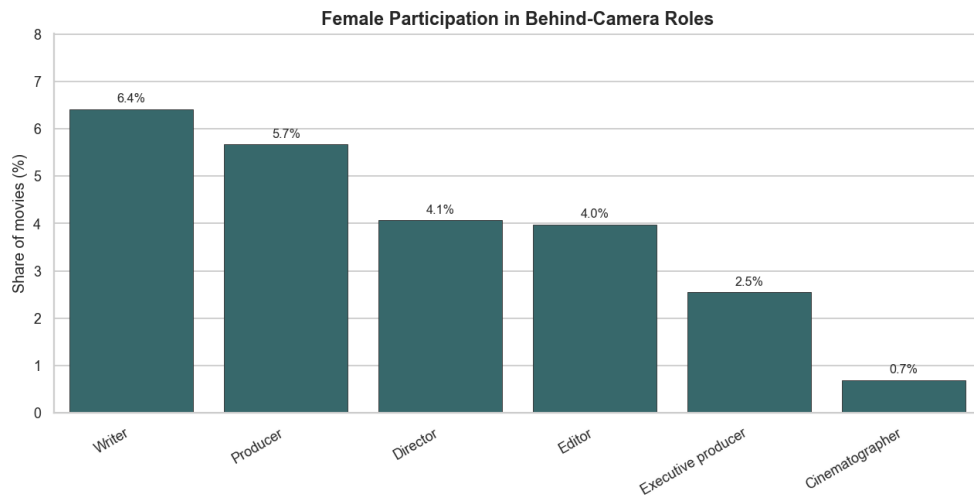


Figure 3.11. Female participation in selected behind-camera roles.

The notebook distributions confirm the same issue from another angle: most movies have very low female crew representation, and the creative leadership score is concentrated near zero. Movies with substantial female creative leadership are therefore uncommon in the dataset.

Female-directed movies also differ strongly in other creative roles. Compared with movies without female directors, female-directed movies are much more likely to have female writers, producers, editors, and cinematographers. The writer gap is especially large at about 45.8 percentage points, followed by producer at 14.5 points, editor at 8.7 points, and cinematographer at 3.2 points.

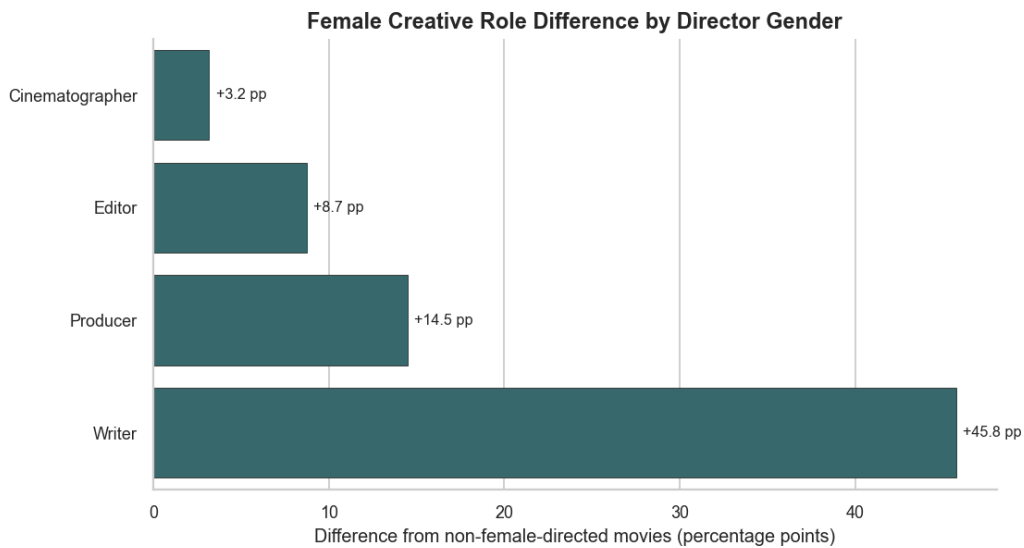


Figure 3.12. Female creative role difference by director gender.

Female-directed movies therefore appear to be part of broader creative networks where women are more present across multiple roles. Representation is not only a director-level pattern; it can cluster within the production team.

The relationship between female crew ratio and ratings is weakly negative in the EDA scatterplots. The Spearman correlations in the notebook are small for TMDB, IMDb, and MovieLens

ratings. These results do not show that women in crew roles reduce ratings; rather, they suggest that simple rating averages do not move strongly upward with higher crew representation in the raw data. Representation is clearly related to who appears in movies and who works on them, but rating behavior remains a separate and relatively noisy outcome.

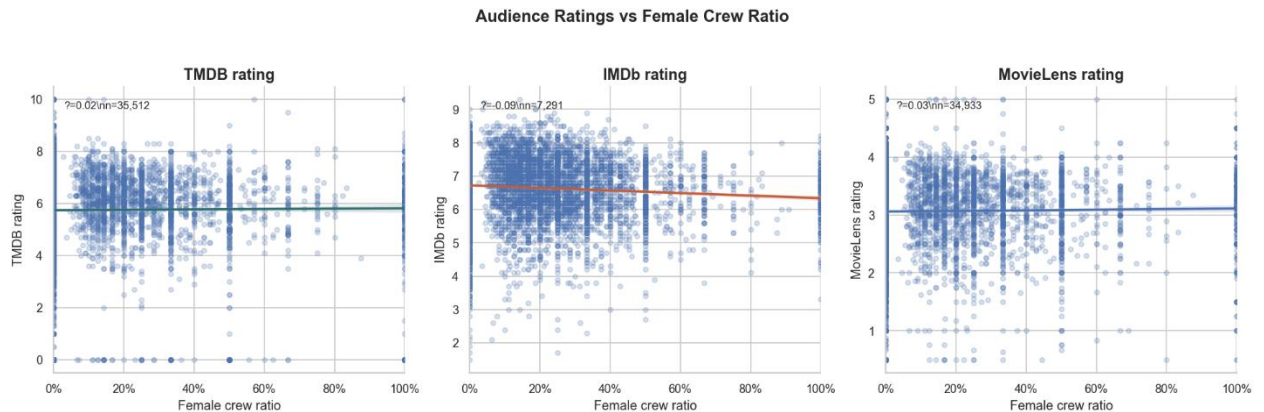


Figure 3.13. Audience ratings versus female crew ratio.

Engagement metrics show a stronger and more positive relationship with female crew ratio. The scatterplots show positive Spearman correlations for log popularity, log TMDB vote count, and log IMDb vote count. These relationships are moderate rather than large, but they are visibly stronger than the rating relationships.

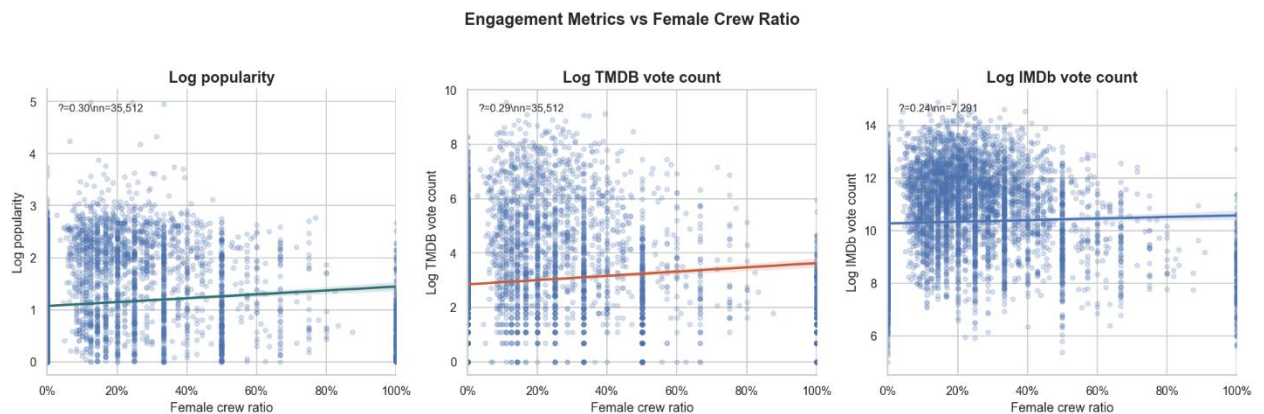


Figure 3.14. Engagement metrics versus female crew ratio.

Female crew ratio has only weak relationships with rating scores, but it has stronger positive relationships with audience attention measures. Crew representation therefore appears more relevant for engagement than for ratings in the raw descriptive patterns.

Financial outcomes show another mixed but meaningful pattern. Female crew ratio has a positive relationship with log revenue and profit in the plotted data, while ROI is almost flat. Revenue and profit appear more responsive to crew representation than ROI.

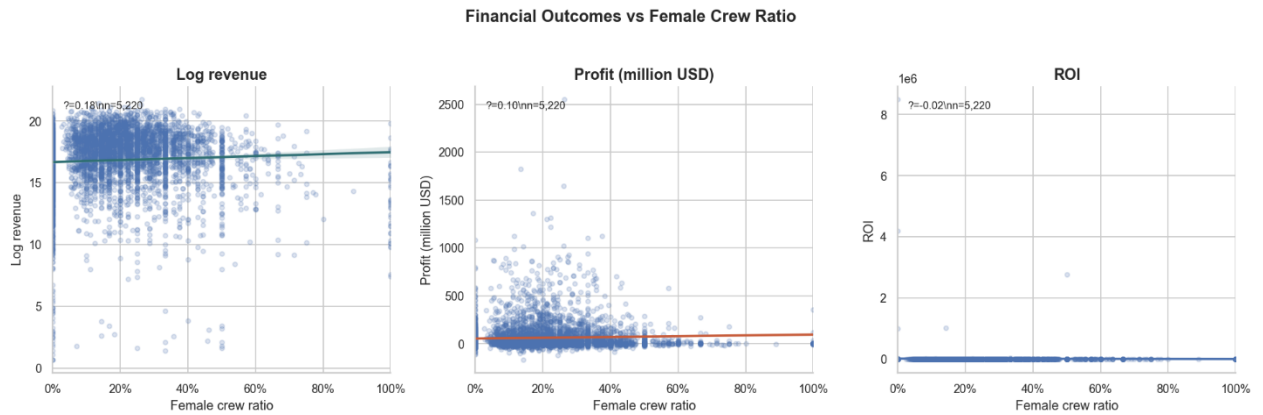


Figure 3.15. Financial outcomes versus female crew ratio.

Revenue, profit, and ROI capture different financial ideas. Revenue reflects market size, profit accounts for budget, and ROI scales return relative to cost. A representation variable can relate to one financial outcome without showing the same pattern for the others.

The writer and producer comparison gives a role-level view of creative participation. Movies with female producers have especially higher log popularity and log revenue, and both female writer and producer status are associated with positive differences in TMDB rating, log popularity, and log revenue. IMDb rating differences are slightly negative.

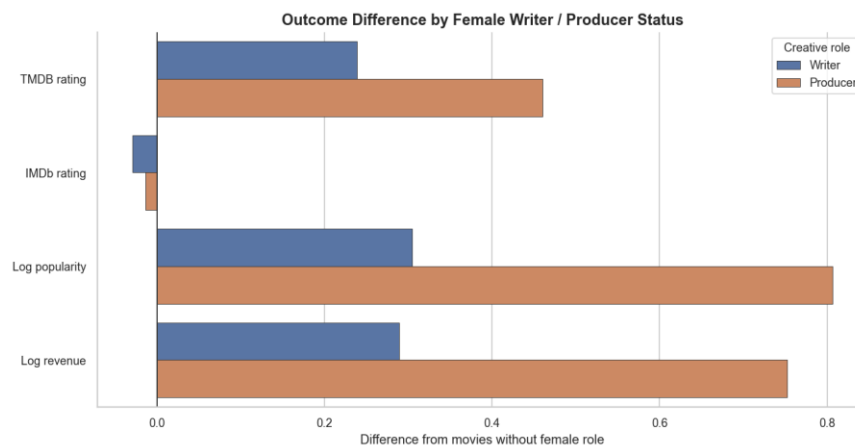


Figure 3.16. Outcome differences by female writer and producer status.

The result shows that not all behind-camera roles behave the same way. Producer status appears particularly connected to engagement and revenue, which is reasonable because producers are tied to financing, packaging, distribution access, and overall project scale. Writer status is more directly connected to story and character representation.

3.4. Linking Behind Camera and On-Screen Representation

The scatterplot between female crew ratio and top-5 female cast share shows a positive relationship. The data are discrete on the cast side because top-5 cast share moves in fifths, but the upward trend is still visible.

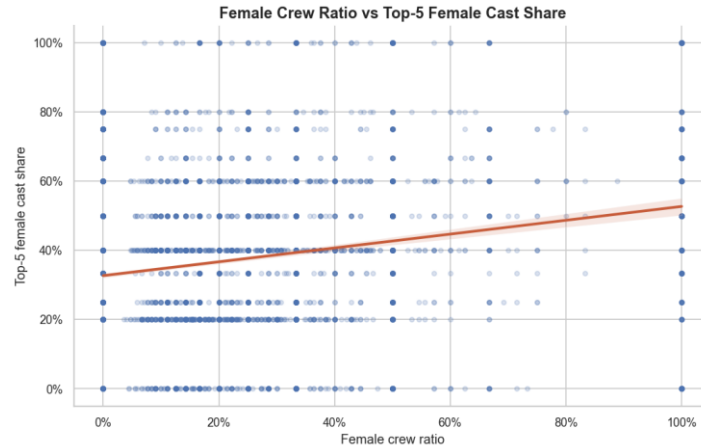


Figure 3.17. Female crew ratio versus top-5 female cast share.

Movies with more women in crew positions are also more likely to have women in central cast positions. The two forms of representation appear connected in the cleaned dataset.

3.5. Bechdel Result and Genre Context

The Bechdel result provides a story-based representation measure. Movies that pass the Bechdel test have a much higher top-5 female cast share than movies that do not pass. However, passing the Bechdel test is not automatically associated with higher IMDb rating or higher log popularity in the descriptive comparison. Representation is clearly visible in the Bechdel comparison, but the success outcomes remain mixed.

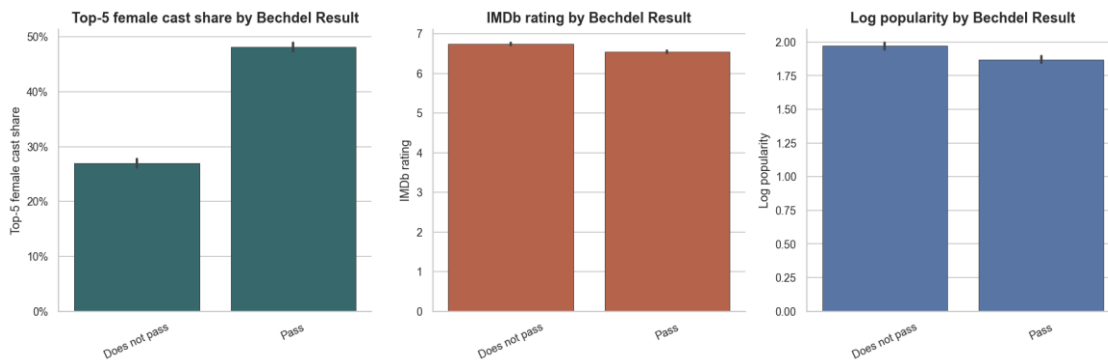


Figure 3.18. Representation and outcomes by Bechdel result.

The genre summary compares three representation measures: top-5 female cast share, female crew ratio, and creative leadership score. Romance, Horror, and Drama are high in top-5 female cast share; Documentary is high in female crew ratio but low in central cast share; and Drama has the highest creative leadership score. A genre can have more women on screen without having more women behind the camera, or the reverse.

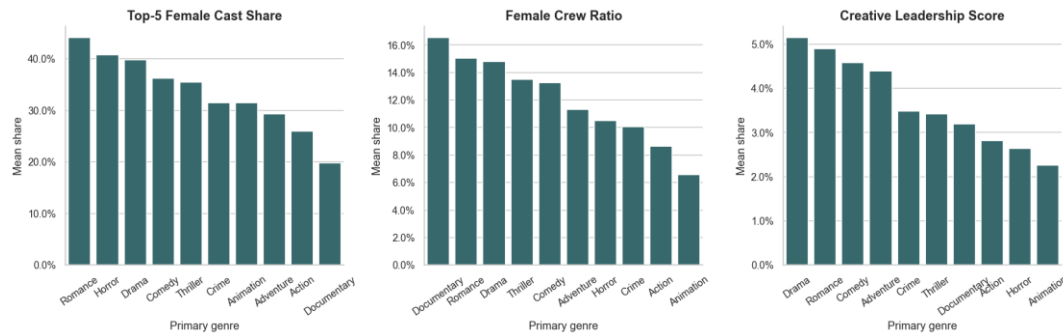


Figure 3.19. Representation measures by primary genre.

Overall, female representation is uneven but patterned. Women are underrepresented in directing and behind-camera roles, and creative leadership is concentrated near zero for most movies. At the same time, movies with female directors, female writers, female producers, and higher female crew ratios tend to show stronger on-screen representation.

The relationship between representation and success is more complicated than a single positive or negative trend. Ratings show weak or mixed descriptive patterns, while engagement and some financial outcomes show clearer relationships, especially for crew representation. The main descriptive finding is that representation is multidimensional: cast, crew, creative leadership, genre, and success outcomes all interact in different ways.

4. Hypothesis Testing

Hypothesis testing was used to examine whether the patterns observed in the exploratory analysis are statistically supported in the cleaned movie dataset. The tests follow the same outcome structure used in the project: ratings, audience engagement, and financial performance. These outcomes are kept separate because they capture different forms of success. A movie may attract a large audience without receiving higher ratings, and strong revenue does not always imply a high return on investment.

The results are interpreted as associations rather than causal effects. The dataset is observational, and movie outcomes are also influenced by genre, release period, budget scale, country, marketing, franchise status, and other production factors. The tests therefore show whether female representation variables are related to outcomes in this dataset, not whether representation alone causes those outcomes.

For binary representation variables, including female director, writer, and producer status, Welch's t-test is used to compare group means. Mann-Whitney U tests are also reported because several movie outcomes are skewed, especially popularity, vote counts, revenue, profit, and ROI. For continuous representation measures, such as top-5 female cast share and female crew ratio, Spearman correlation is used to measure monotonic relationships without assuming a linear pattern.

4.1. Hypothesis 1: Director Gender and Movie Outcomes

The first hypothesis compares movies with at least one female director to movies without a female director. All three rating comparisons are statistically significant, but the direction is not consistent across platforms. Female-directed movies have higher mean TMDB and MovieLens ratings, while movies without a female director have a higher mean IMDb rating.

The engagement results are also mixed. Female-directed movies have higher log popularity and higher log TMDB vote count, but lower log IMDb vote count. In the financial subset, female-directed movies have significantly lower log revenue. Profit and ROI are less stable: Welch's test does not find significant mean differences, while the Mann-Whitney test suggests distributional differences. Because financial outcomes are highly skewed, these results should be read cautiously.

Outcome	n female	n non-female	Female mean	Non-female mean	Difference	Welch p	M-W p	Welch result
TMDB rating	1,844	43,502	5.772	5.618	+0.154	0.0002	0.0274	Significant
IMDb rating	508	7,201	6.447	6.653	-0.206	<0.001	<0.001	Significant
MovieLens rating	1,837	42,789	3.133	3.057	+0.076	<0.001	0.0004	Significant
Log popularity	1,844	43,502	1.065	0.977	+0.088	<0.001	<0.001	Significant
Log TMDB vote count	1,844	43,502	2.810	2.651	+0.159	0.0001	<0.001	Significant
Log IMDb vote count	508	7,201	9.601	10.249	-0.648	<0.001	<0.001	Significant
Log revenue	231	5,144	16.184	16.768	-0.584	0.0029	0.0004	Significant
Profit (million USD)	231	5,144	51.541	59.632	-8.091	0.3728	0.0089	Not significant
ROI	231	5,144	3.390	5,822.182	-5,818.793	0.0599	0.0066	Not significant

Table 4.1. Director gender group comparisons across rating, engagement, and financial outcomes.

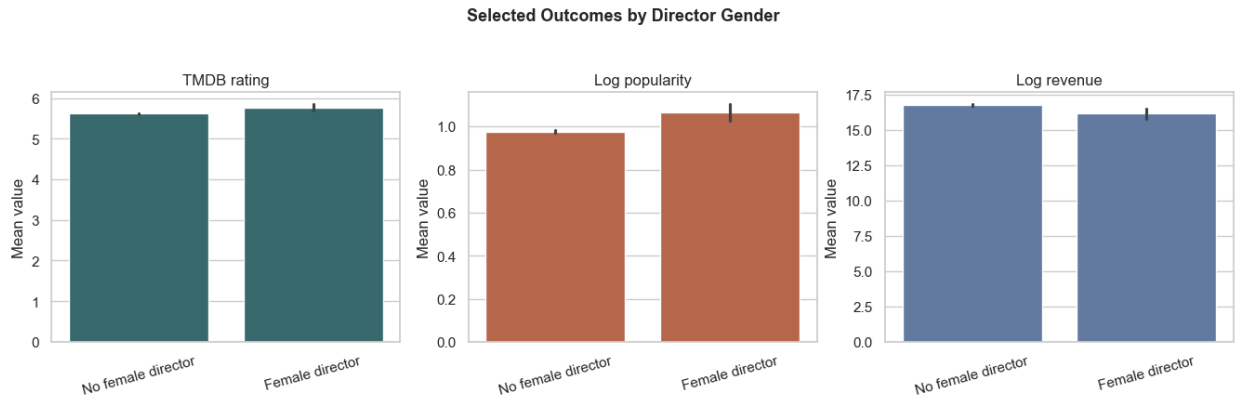


Figure 4.1. Selected outcomes by director gender.

Taken together, the director-gender results show that female-directed movies cannot be described as simply more or less successful overall. The rating outcomes are statistically significant, but they point in different directions: female-directed movies have slightly higher TMDB and MovieLens ratings, while the IMDb average is lower. Engagement follows the same mixed pattern. Female-directed movies receive higher TMDB-based popularity and vote-count values, but they have lower IMDb vote counts, which may reflect differences in audience base, platform coverage, or the types of films represented in the IMDb subset.

Financial outcomes are also mixed. Log revenue is lower for female-directed movies, while profit and ROI are not significant under Welch's test. The Mann-Whitney results suggest distributional differences for profit and ROI, which is reasonable given how skewed financial variables are. Overall, director gender is related to several outcomes, but the evidence does not support one clear success direction across ratings, engagement, and financial performance.

4.2. Hypothesis 2: Top-5 Female Cast Share and Movie Outcomes

The second hypothesis tests whether on-screen female representation is associated with movie outcomes. Top-5 female cast share is used because it focuses on the most visible billed cast members rather than the full cast list. Most correlations are statistically significant, but the effect sizes are small, which is important given the large sample sizes.

For ratings, all three correlations are negative and weak. IMDb rating has the largest negative correlation among the rating outcomes, with $\rho = -0.1287$. Engagement results are less consistent: log popularity has a very small positive relationship, TMDB vote count is not statistically significant, and IMDb vote count has a stronger negative relationship. Financial outcomes are also weak. Log revenue and profit have small negative correlations, while ROI is not significant.

Outcome	n	Spearman rho	p-value	Result
TMDB rating	39,677	-0.0595	<0.001	Significant
IMDb rating	7,565	-0.1287	<0.001	Significant
MovieLens rating	39,049	-0.0393	<0.001	Significant
Log popularity	39,677	+0.0224	<0.001	Significant
Log TMDB vote count	39,677	+0.0061	0.2255	Not significant
Log IMDb vote count	7,565	-0.1943	<0.001	Significant
Log revenue	5,320	-0.0572	<0.001	Significant
Profit (million USD)	5,320	-0.0284	0.0386	Significant
ROI	5,320	-0.0080	0.5617	Not significant

Table 4.2. Spearman correlations between top-5 female cast share and movie outcomes.

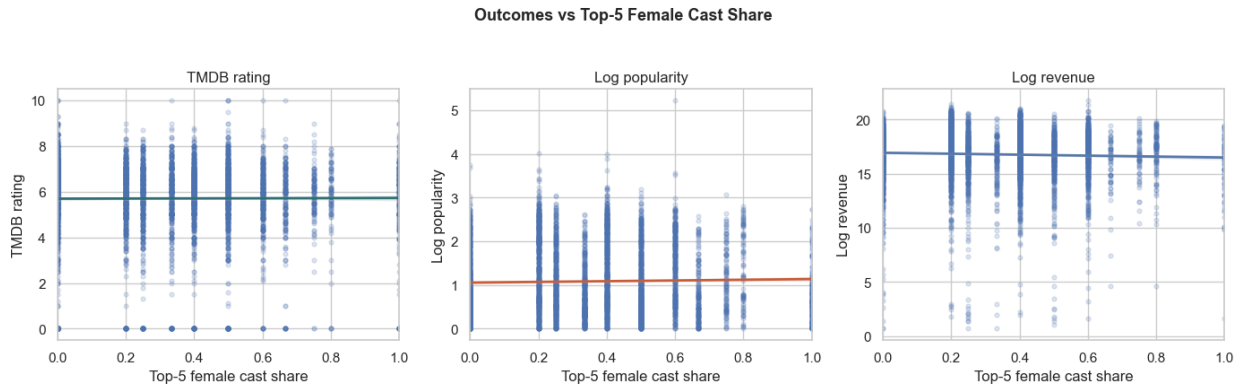


Figure 4.2. Selected outcomes versus top-5 female cast share.

Overall, top-5 female cast share is statistically related to several outcomes, but the relationships are small and not consistently positive. The results suggest that on-screen representation is associated with movie outcomes, but the direction and strength of the relationship depend on the type of success being measured.

4.3. Hypothesis 3: Female Crew Ratio and Movie Outcomes

The third hypothesis tests female crew ratio, which summarizes women's participation across recorded crew roles. Compared with top-5 female cast share, female crew ratio has a clearer relationship with engagement and some financial outcomes. Rating correlations are statistically significant but weak: TMDB and MovieLens are slightly positive, while IMDb is slightly negative.

The strongest results appear in engagement outcomes. Female crew ratio has positive correlations with log popularity, log TMDB vote count, and log IMDb vote count. The correlation values are around

0.24 to 0.30, which are noticeably larger than the cast-share correlations. Financial outcomes also show positive associations with log revenue and profit, while ROI is not significant. This indicates that crew representation is more closely related to audience reach and revenue scale than to rating averages or ROI.

Outcome	n	Spearman rho	p-value	Result
TMDB rating	35,512	+0.0245	<0.001	Significant
IMDb rating	7,291	-0.0920	<0.001	Significant
MovieLens rating	34,933	+0.0307	<0.001	Significant
Log popularity	35,512	+0.2957	<0.001	Significant
Log TMDB vote count	35,512	+0.2883	<0.001	Significant
Log IMDb vote count	7,291	+0.2355	<0.001	Significant
Log revenue	5,220	+0.1767	<0.001	Significant
Profit (million USD)	5,220	+0.0968	<0.001	Significant
ROI	5,220	-0.0166	0.2311	Not significant

Table 4.3. Spearman correlations between female crew ratio and movie outcomes.

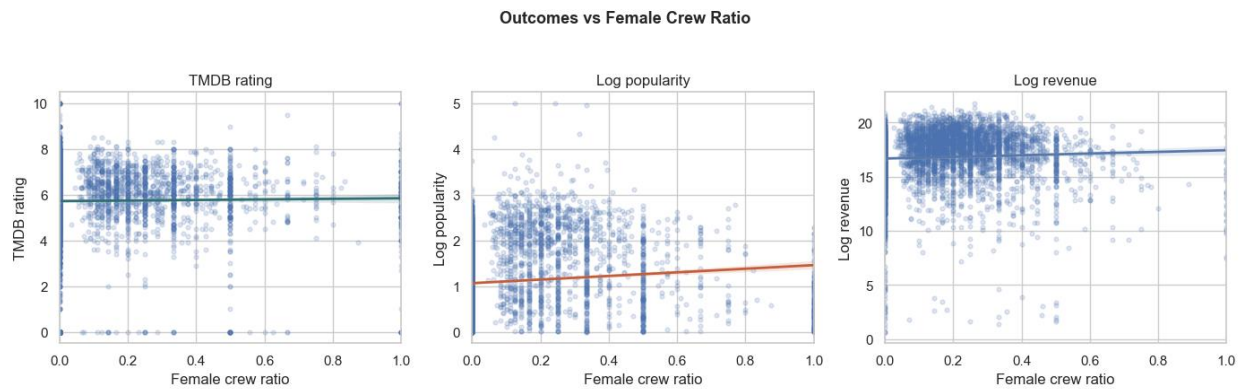


Figure 4.3. Selected outcomes versus female crew ratio.

The female crew ratio results provide the strongest success-related evidence in this section. Unlike top-5 female cast share, which has mostly small and inconsistent relationships, female crew ratio is positively associated with all three engagement measures. The correlations with log popularity, log TMDB vote count, and log IMDb vote count are moderate compared with the other representation tests, suggesting that movies with higher female crew participation tend to receive more audience attention.

The financial results also support this pattern: female crew ratio is positively associated with log revenue and profit, although ROI remains not significant. This distinction is important because revenue and profit reflect the scale of market performance, while ROI depends heavily on production cost and can be distorted by extreme values. Rating outcomes remain weak, so the main contribution of female crew representation appears to be connected more to audience reach and financial scale than to higher average ratings.

4.4.Hypothesis 4: Creative Roles and On-Screen Representation

The final hypothesis group connects behind-camera creative roles with on-screen representation. Movies with a female director, female writer, or female producer all have significantly higher top-5 female cast share than movies without women in those roles. The largest differences are found for female directors and female writers, two roles that are closely tied to story perspective, casting priorities, and creative direction.

Role comparison	n with role	n without role	Mean with role	Mean without role	Difference	Welch p
Female director vs top-5 female cast share	1,628	38,049	0.493	0.345	+0.149	<0.001
Female writer vs top-5 female cast share	2,801	36,876	0.477	0.341	+0.136	<0.001
Female producer vs top-5 female cast share	2,486	37,191	0.398	0.348	+0.050	<0.001

Table 4.4. Female creative roles and top-5 female cast share.

The chi-square test between female director status and female lead status is also significant: chi-square = 761.3896, $df = 1$, $p < 0.001$. In the contingency table, 868 out of 1,844 female-directed movies have a female lead, compared with 8,783 out of 43,501 movies without a female director. This result does not test success directly, but it helps explain how behind-camera representation connects to the representation variables used in the outcome analyses.

4.5. Hypothesis Testing Summary

The hypothesis tests support three main conclusions. First, movies with female directors differ from other movies, but the direction of the difference depends on the outcome. Ratings are mixed across platforms, engagement differs by vote source, and financial results are sensitive to skewed distributions. Second, top-5 female cast share has statistically significant relationships with several outcomes, but the effects are generally small and not consistently positive. Third, female crew ratio shows the strongest and most consistent association with engagement and selected financial outcomes.

Overall, the strongest success-related evidence appears in engagement and selected financial outcomes, especially for female crew ratio. Rating outcomes are weaker and less consistent. These results support the main argument of the project: female representation is connected to movie success, but the relationship depends on how success is measured.

5. Machine Learning

The machine learning analysis extends the earlier descriptive and statistical results by asking whether representation variables help predict movie success after basic movie characteristics are already included. The models include controls such as release year, genre, budget, runtime, language, and production country, then compare them with models that add cast, crew, and Bechdel-related variables. The aim is predictive rather than causal: the section evaluates whether representation carries useful information for different success outcomes.

Success is kept separate across the models instead of being reduced to one overall score. IMDb rating represents audience evaluation, profitability represents financial performance, and popularity represents audience attention. This structure matters for the project because representation may be connected to visibility or revenue even when it does not improve average ratings.

Feature group	Main variables included	Purpose in the model
Controls only	Release year, decade, budget, budget missing flag, runtime, genre, language, production country	Baseline model using general movie information

Feature group	Main variables included	Purpose in the model
Controls + cast	Baseline controls plus female cast ratios, top-billed female share, female lead, and second-billed female indicator	Tests whether on-screen representation improves prediction
Controls + crew	Baseline controls plus female director, writer, producer, editor, cinematographer, crew ratio, and creative leadership score	Tests whether behind-camera representation improves prediction
Controls + cast + crew + Bechdel	All control, cast, crew, Bechdel pass, and Bechdel rating variables	Full representation model

Table 5.1. Feature groups used in the machine learning models.

The same preprocessing approach is used across the models. Numeric variables are median-imputed and scaled, while categorical variables are imputed and one-hot encoded. The data is split into training and test sets using a 75/25 division. Classification performance is compared mainly with ROC-AUC, with F1 score reported as a supporting metric. Logistic Regression provides a simpler benchmark, while Random Forest allows non-linear relationships between movie characteristics and success outcomes.

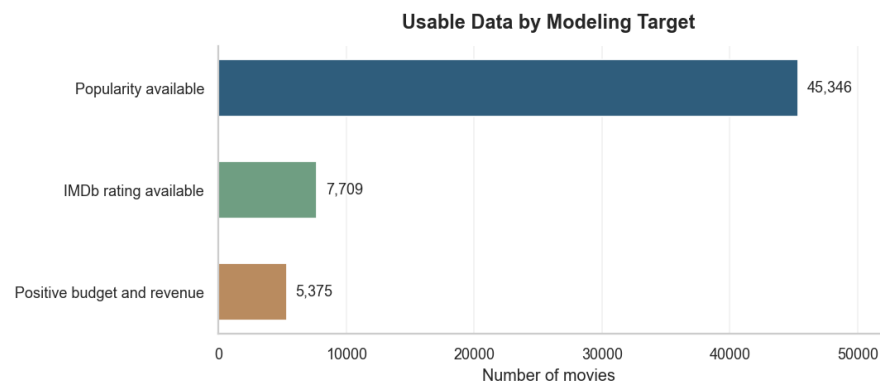


Figure 5.1. Usable data by modeling target.

Modeling target	Definition	Rows used	Success class / outcome
High IMDb rating	IMDb rating at or above 6.70	7,709	Binary classification
Profitability	Revenue greater than budget	5,375	Binary classification
High popularity	Popularity at or above the median value of 1.13	45,346	Binary classification
IMDb rating regression	Continuous IMDb average rating	7,709	Regression check

Table 5.2. Prediction targets and sample availability.

5.1. Predicting High IMDb Rating

High IMDb rating is used as the rating-based success outcome. It reflects how viewers evaluate a movie on average, rather than how widely the movie reaches audiences or how much money it makes. The

strongest result for this outcome comes from the Random Forest model with controls only, with a ROC-AUC of about 0.816 and an F1 score of 0.745. Adding cast variables, crew variables, or the full representation set does not improve the best score.

This result is consistent with the earlier rating findings. Representation variables showed some statistically significant relationships with ratings, but the direction and size of those relationships were not stable across platforms. In the prediction setting, the model relies more on general movie information such as runtime, genre, year, language, and budget. For rating success, representation does not add much beyond these controls, which suggests that audience evaluation is not the main outcome where representation contributes predictive signal.

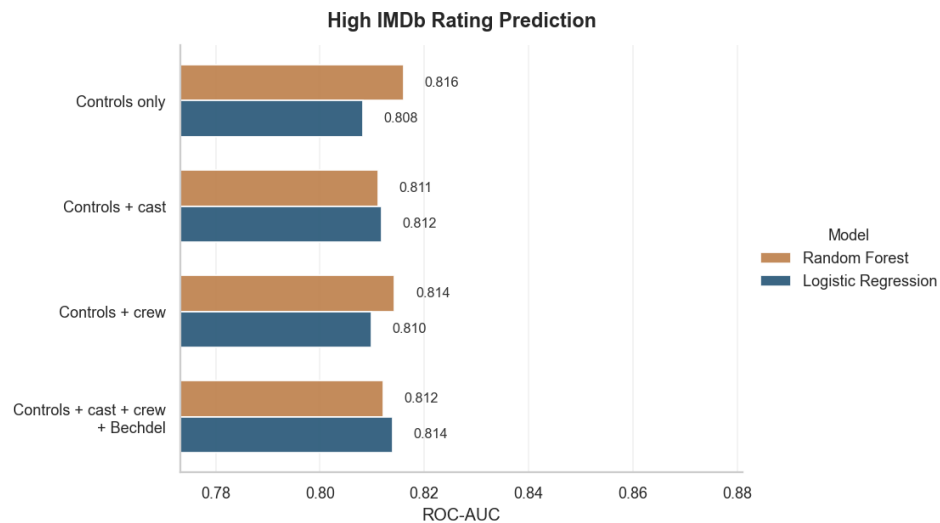


Figure 5.2. ROC-AUC results for high IMDb rating prediction.

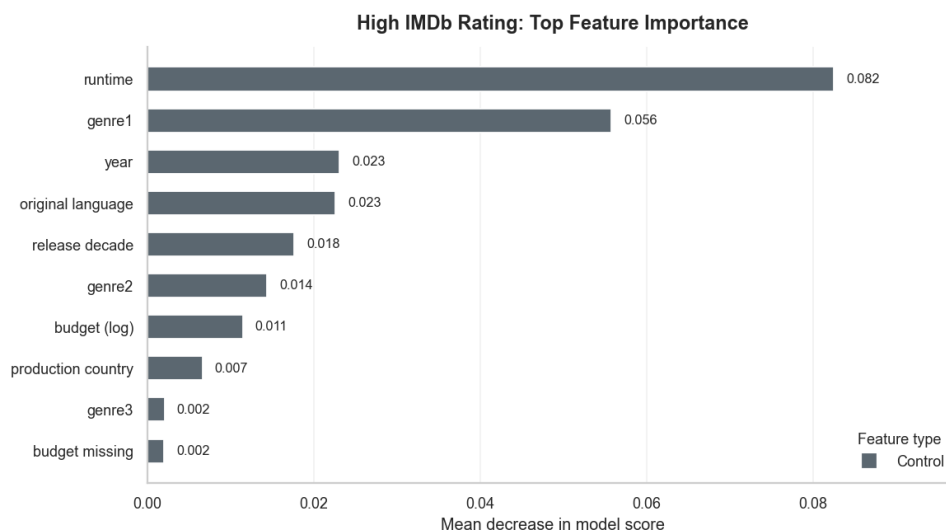


Figure 5.3. Most important variables in the best high IMDb rating model.

To check whether this result depends on turning IMDb rating into a high-rating label, the notebook also tests IMDb rating as a continuous outcome. In this regression version, the model predicts the actual

IMDb score instead of classifying movies as high-rated or not. The full Random Forest model performs slightly better than the controls-only Random Forest model, with an R-squared value of about 0.370 compared with 0.362. However, this improvement is very small.

This supports the same conclusion as the classification model. Representation variables may explain a limited amount of variation in IMDb rating, but they do not strongly improve rating prediction once standard movie controls are included. Compared with the gains seen later for profitability and popularity, the rating result remains weak. Overall, rating success appears less responsive to representation features than engagement or financial success.

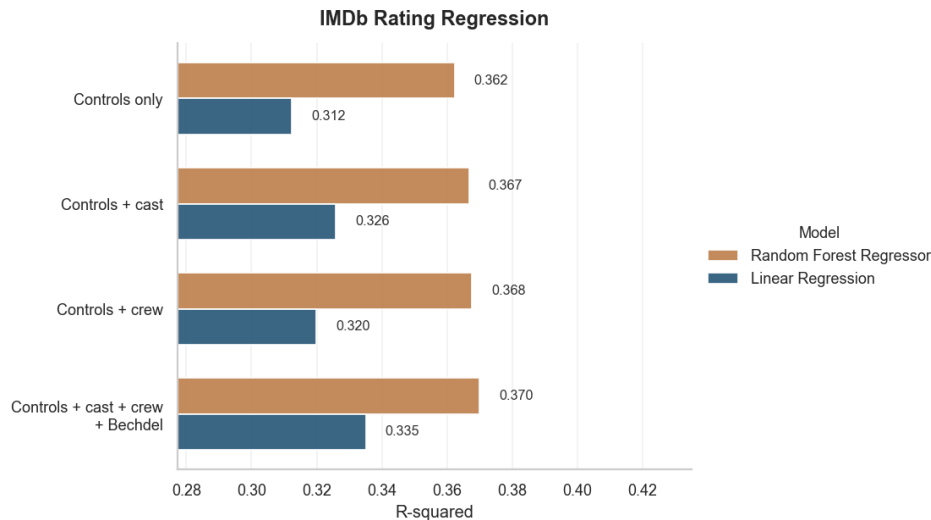


Figure 5.4. R-squared results for IMDb rating regression.

Regression task	Best model	Feature group	MAE	RMSE	R-squared
IMDb rating	Random Forest Regressor	Controls + cast + crew + Bechdel	0.578	0.752	0.370
IMDb rating baseline	Random Forest Regressor	Controls only	0.582	0.757	0.362

Table 5.3. IMDb rating regression check.

5.2. Predicting Profitability

Profitability is used as the financial success outcome. A movie is classified as profitable when its revenue is greater than its budget. Unlike IMDb rating, this outcome is about market performance: whether a film earns back more than it costs. The strongest model is the Random Forest using controls, cast variables, crew variables, and Bechdel variables, with a ROC-AUC of about 0.744 and an F1 score of 0.793. Compared with the best controls-only Random Forest baseline, the full representation model improves ROC-AUC by about 0.054. This is a larger improvement than the changes observed in the IMDb rating task, so the financial model gives stronger evidence that representation variables matter for one dimension of success.

This is one of the clearest places where representation contributes to success prediction. The model is not claiming that representation causes profitability, but it does show that representation variables help separate profitable movies from non-profitable ones when they are combined with standard

controls. The feature-importance plot supports this interpretation: Bechdel rating, Bechdel pass, and female crew ratio appear alongside budget, year, genre, and production context. This means the model is not relying only on financial scale or production characteristics; some representation-related information remains useful after those factors are included. Financial performance therefore seems to capture a success dimension where representation is more informative than it is for average rating.

Because this model uses movies with positive budget and revenue values, the result should be interpreted specifically as a financial-performance finding. Even with that boundary, the pattern is clear: representation variables add more predictive value for profitability than they do for high IMDb rating.

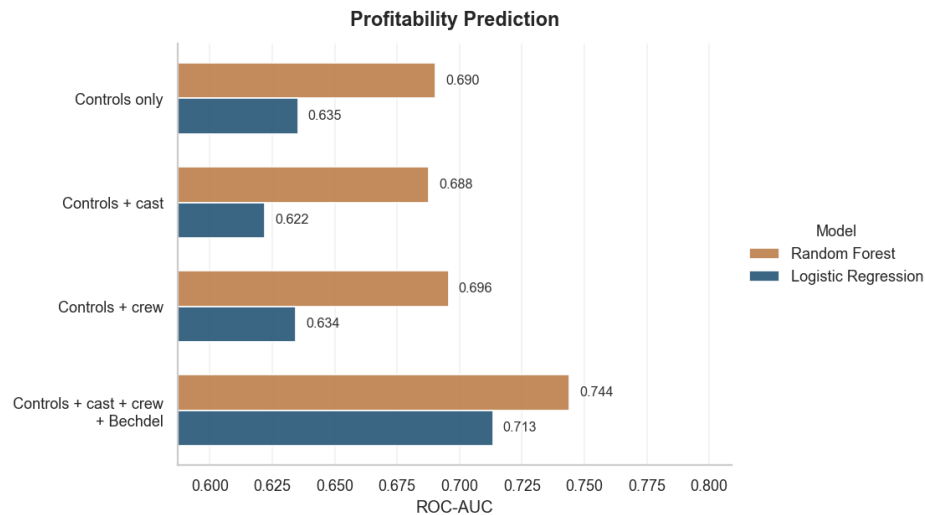


Figure 5.5. ROC-AUC results for profitability prediction.

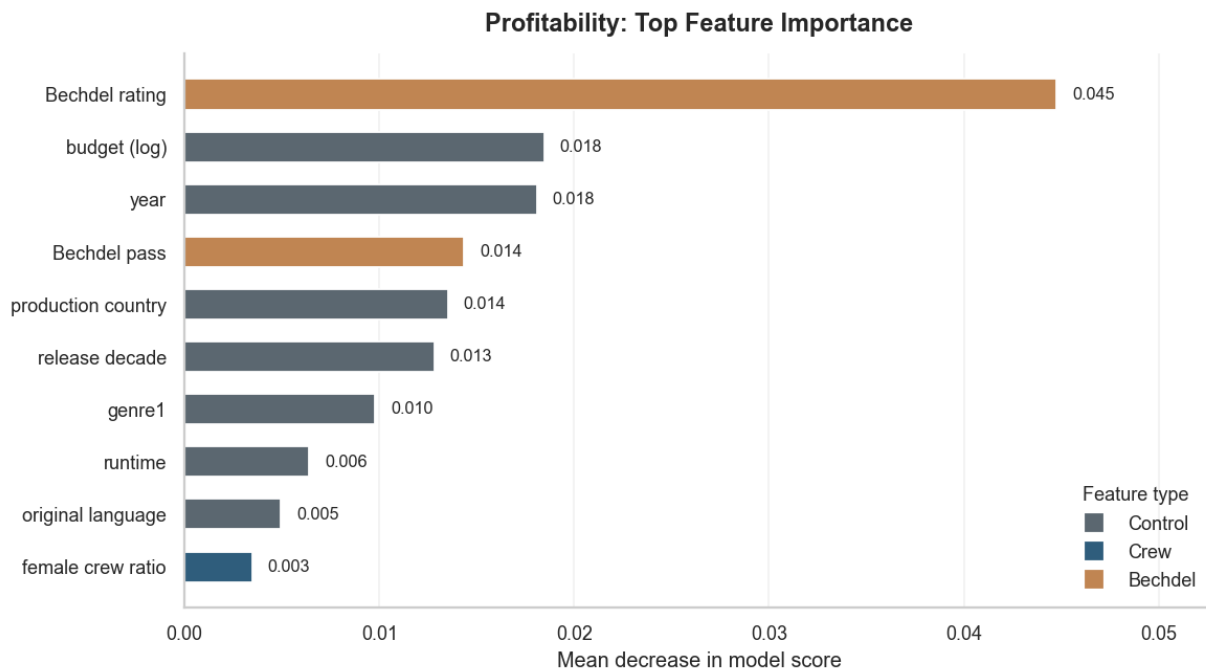


Figure 5.6. Most important variables in the best profitability model.

5.3. Predicting High Popularity

High popularity is used as the engagement-based success outcome. It measures attention and reach rather than viewer evaluation. The strongest model is Logistic Regression with the full representation feature set, reaching a ROC-AUC of about 0.845 and an F1 score of 0.748. This is the strongest classification result in the machine learning section. The full model performs better than the controls-only model, which suggests that representation variables help predict whether a movie becomes more visible or widely discussed.

The popularity model gives the strongest link between representation and success. Controls-only models already perform well, but adding representation variables increases ROC-AUC by about 0.046. This matches the hypothesis-testing results, where female crew ratio had stronger relationships with engagement metrics than with rating outcomes. The feature-importance plot also places female crew ratio and female cast share among the useful predictors, together with year, genre, budget, and Bechdel variables. For this project, that is an important result: representation is more useful for predicting whether a movie receives attention than for predicting whether it receives a high average rating. Popularity therefore behaves more like a visibility outcome than a quality judgment, and representation seems more connected to that visibility dimension.

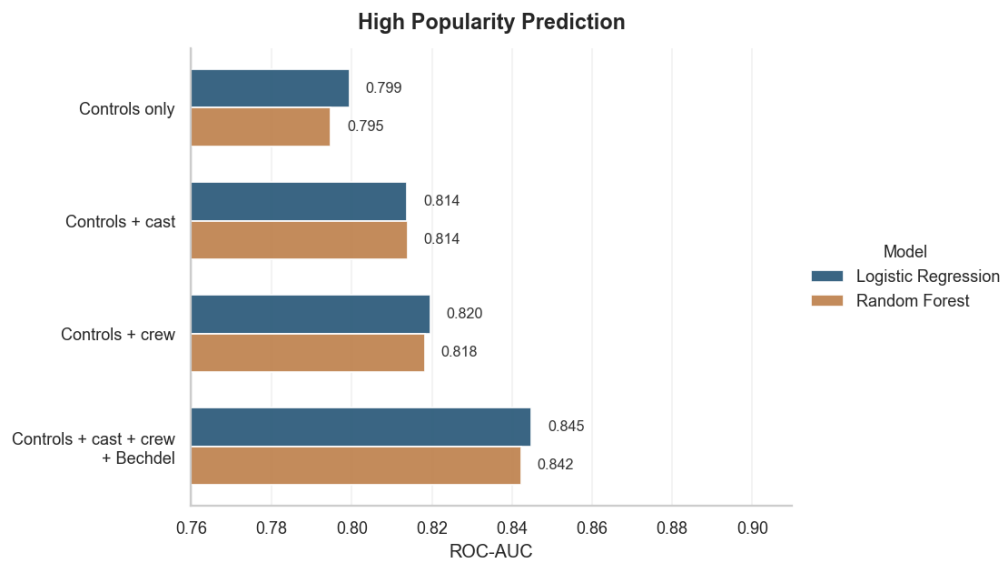


Figure 5.7. ROC-AUC results for high popularity prediction.

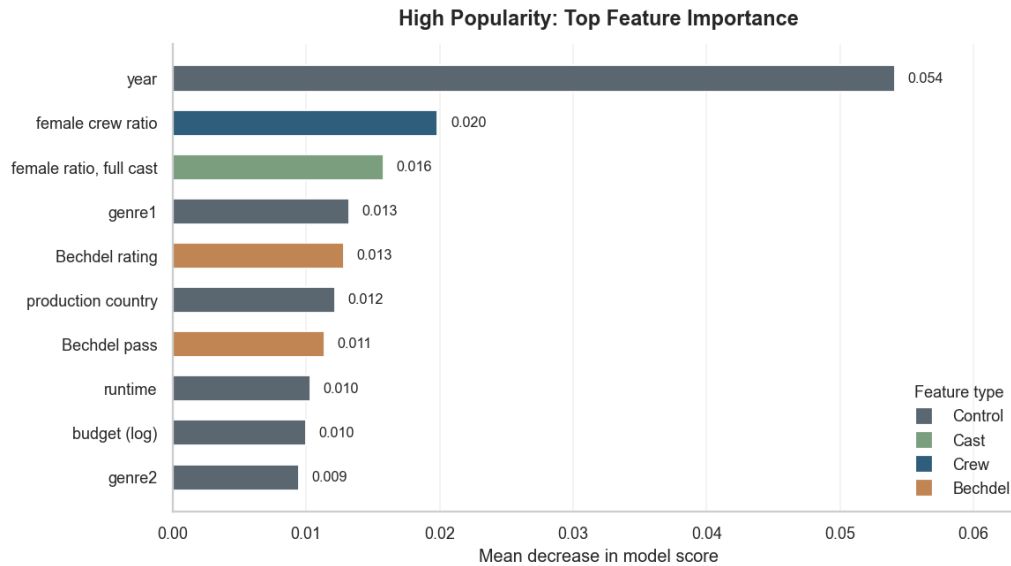


Figure 5.8. Most important variables in the best high popularity model.

Success outcome	Best model	Best feature group	ROC-AUC	F1	Gain over controls
High IMDb rating	Random Forest	Controls only	0.816	0.745	+0.000
Profitability	Random Forest	Controls + cast + crew + Bechdel	0.744	0.793	+0.054
High popularity	Logistic Regression	Controls + cast + crew + Bechdel	0.845	0.748	+0.046

Table 5.4. Best classification models by success target.

5.4. Added Predictive Value of Representation

The added-value comparison gives the most direct machine learning answer to the project objective. Representation variables do not improve high IMDb rating prediction, but they improve profitability and high popularity prediction. The improvement is largest for profitability and also clear for popularity. This split is important because it shows why the report does not treat success as one combined measure: the same feature group can be weak for rating success but useful for engagement and financial success.

The cast-versus-crew breakdown also matters for interpretation. Cast variables alone do not explain the full gain. Crew variables add more value for popularity, and the strongest results come from combining cast, crew, and Bechdel features. This connects the machine learning results back to the EDA and hypothesis testing sections: on-screen representation is visible in the descriptive patterns, but behind-camera participation is more consistently connected to engagement and financial outcomes. The full representation model works best when it captures more than one layer of representation, including who appears on screen, who works behind the camera, and whether the film passes the Bechdel measure.

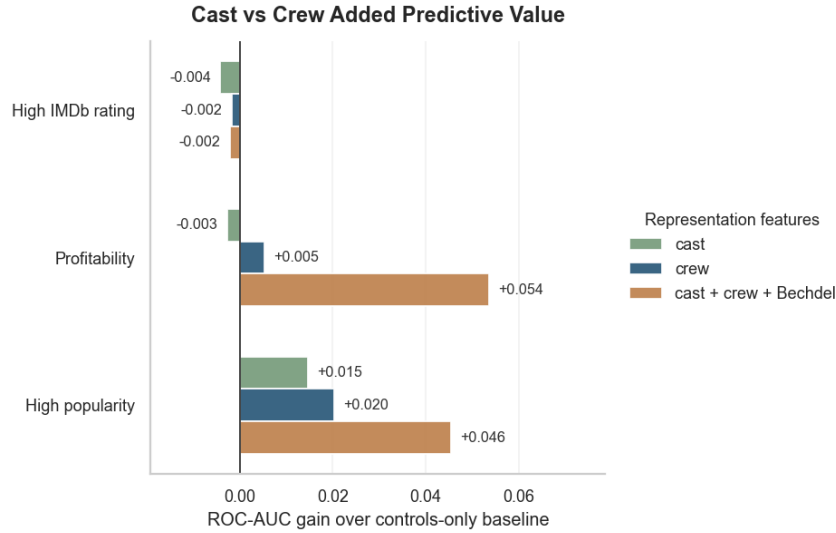


Figure 5.9. Added predictive value of cast and crew representation feature groups.

Outcome	Cast features	Crew features	Full representation features	Interpretation
High IMDb rating	-0.004	-0.002	-0.002	No predictive improvement over controls
Profitability	-0.003	+0.005	+0.054	Full representation set improves financial prediction
High popularity	+0.015	+0.020	+0.046	Representation improves engagement prediction

Table 5.5. Added ROC-AUC value relative to the controls-only baseline.

5.5. Machine Learning Summary

The machine learning results strengthen the main finding of the project. Female representation is not linked to success in one simple direction. For ratings, representation does not add meaningful predictive value once standard movie controls are included. For engagement and financial performance, the representation variables provide clearer predictive information, especially when crew and Bechdel features are included with cast features. This pattern is consistent across the classification tasks and the regression check, which makes the conclusion stronger than relying on one model alone.

The strongest machine learning evidence therefore connects representation to visibility and market performance rather than to average rating alone. Movies with stronger female representation are not automatically rated higher, but representation helps predict whether movies reach larger audiences and whether they are financially successful. This supports the report's central point: success has to be separated into ratings, engagement, and financial performance before the relationship with representation becomes clear.

6. Findings

The results show that female representation in movies is connected to movie success, but the relationship depends on how success is defined. The project does not support a simple claim that movies with stronger female representation are always more or less successful. Instead, representation is more clearly related to audience engagement and financial performance than to average audience ratings.

Female representation is uneven across the dataset. Female directors make up only a small share of movies, and women are also underrepresented in several behind-camera roles. At the same time, representation tends to cluster: movies with female directors or women in creative roles usually have higher female cast representation and higher female crew participation. This suggests that on-screen and behind-camera representation are closely linked.

For rating-based success, the findings are mixed. Female-directed movies have slightly higher TMDB and MovieLens ratings, while movies without female directors have a higher average IMDb rating. The machine learning results confirm this pattern: representation variables do not improve high IMDb rating prediction beyond the controls-only model, and the IMDb regression check shows only a very small improvement. This means representation is not a strong predictor of rating success in this project.

The results are stronger for audience engagement. Female crew ratio has positive relationships with popularity and vote-count measures, and representation variables improve high popularity prediction in the machine learning models. This suggests that representation is more connected to visibility and audience reach than to average rating alone.

Financial performance also shows a clearer connection with representation than ratings do. The profitability model performs best when cast, crew, and Bechdel variables are added to the controls. This does not prove causality, but it shows that representation helps distinguish profitable movies from non-profitable ones.

Overall, crew representation appears especially important. Female crew ratio performs more strongly than top-5 female cast share in several tests and appears as a useful feature in the machine learning models. The main result is that female representation is associated with success, but not equally across all measures: ratings are weaker and inconsistent, while engagement and profitability show clearer links.

7. Limitations and Future Work

The findings should be interpreted within the limits of the available movie data. Movie success is shaped by many factors outside the dataset, such as marketing budget, distribution strategy, franchise recognition, star power, release timing, and cultural context. These factors can influence ratings, popularity, and revenue independently of gender representation.

Another limitation is that gender representation is measured through structured variables such as cast share, crew ratio, female director status, and Bechdel result. These measures are useful for large-scale analysis, but they cannot fully capture the quality, depth, or narrative importance of female characters and creative contributions. For example, a movie may have women in central cast positions but still represent them in limited or stereotypical ways.

Future work could extend the analysis by including richer context about marketing, distribution, production companies, and star power. It could also use plot summaries, scripts, subtitles, or reviews to study how women are represented in the story, not only whether they appear in the cast or crew. A time-based analysis could also examine whether the relationship between representation and success changes across decades.

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